

YU HIN (THOMAS) LIU

yl13534@nyu.edu | (646) 255-8177

EDUCATION

New York University

Bachelor of Science in Computer Science

New York, NY

June 2029

- GPA: 3.5/4 | Relevant Coursework: Object Oriented Programming, Data Structures & Algorithms, Calculus II

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, C#, JavaScript, HTML/CSS
- **AI/ML Frameworks:** PyTorch, TensorFlow, scikit-learn; Experience with neural networks and ML model development
- **Development:** Backend Development (FastAPI, Flask), REST APIs, Docker, eKuiper, System Development Lifecycle, Git/GitHub
- **Technologies:** MySQL, PostgreSQL, React.js, Tailwind CSS, Applications Integration, Data Analysis (Pandas, NumPy)
- **Core Competencies:** Problem-Solving, Debugging, Quality Assurance & Testing, Technical Documentation

EXPERIENCE

Martinrea International Inc. - Software Developer Intern

Vaughan, ON

May 2026 - Aug. 2026

- Built PULSE, an industrial IoT application that streams machine production data to a central server for real-time monitoring
- Engineered C++ and Python ingestion pipelines using eKuiper for stream processing and PostgreSQL for telemetry storage
- Developed a responsive web dashboard (HTML, Tailwind CSS) to visualize live machine data via REST/JSON APIs
- Containerized services with Docker and managed version control through Git/GitHub for reproducible deployments

University of Toronto - Research Assistant, Leading Tomorrow

Toronto, ON

April 2024 - Oct. 2024

- Assisted Professor Tao Wang in analyzing demographic datasets using Python to examine social trends in education
- Designed surveys and conducted 300+ interviews, generating 500+ responses with thorough documentation and reporting

Codebuds - Co-Founder & Technical Lead

Toronto, ON

May 2023 - June 2024

- Led end-to-end development of organization's website, implementing modular backend systems for course and event management serving 10,000+ users
- Developed and maintained code through Git version control, following release management processes and coding standards
- Collaborated across technical and non-technical teams to translate business requirements into scalable technical solutions

PROJECTS

SkinSense Smart Mirror, NYU Tandon - Hardware and ML Developer

New York, NY

Feb, 2026 - April 2026

- Built a Raspberry Pi 4B-based smart mirror with camera and self-rotation for AI-powered skin analysis
- Designed voltage divider circuits and PCB layouts, as well as 3D-printing model using Autodesk Fusion 360
- Tested computer vision and AI analysis pipelines in Python for real-time skin condition detection

Instrument Detection System - Independent Developer

New York, NY

Nov. 2025 - Present

- Designed and trained PyTorch-based CRNN model for multi-label instrument classification from Mel-spectrogram features
- Developed FastAPI backend services and React frontend, demonstrating full-stack ML application deployment capabilities
- Implemented model evaluation pipeline and maintained reproducible experimentation workflow using Git version control

CREATE Summer Program, University of Toronto - Team Lead

Toronto, ON

July 2024 - Aug. 2024

- Analyzed urban and renewable energy datasets using Python (Pandas, scikit-learn) for data-driven problem-solving
- Executed full data pipeline from collection to modeling, applying systematic engineering thinking and collaboration